



The Note Under the Keypad: A Field Deployment of Automated Rotation Reminders for Shared-House Door Codes

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Abstract. Household and workplace security guidance routinely recommends rotating a shared access code, but says little about what happens to a code between the moment it changes and the moment everyone who needs it has it memorised. We report a fourteen-week field deployment of a lightweight rotation-reminder tool across 42 share houses using a shared keypad door code, comparing a push-notification and text-message nudge condition against each house's existing paper-notice process. A weekly, consented photographic audit of each entry area recorded the day a new code first appeared as legible informal notation — a sticky note, a whiteboard scrawl, a taped index card — rather than staying in residents' memory or the official channel. Median time to first informal notation was 2.6 days under the reminder tool and 2.9 days under the standard notice, a difference that did not reach significance; doubling notification frequency inside the reminder-tool arm changed the median by a matter of hours. Self-reported notification recall was high (88%) and uncorrelated with how quickly a house's code appeared on paper. We interpret the null result as evidence that a code's informal legibility is set by household coordination needs rather than by how often, or how insistently, residents are told to remember it themselves.

Introduction

Guidance on a shared access credential is consistent on one point: rotate it on a schedule, and rotate it again whenever someone with access moves on. Two decades of workplace security research have documented what a rotation schedule does not, on its own, guarantee — that the people who need the new credential actually hold it in memory rather than on whatever surface is fastest to write it on. Policies that treat a credential's rotation as the safety event, independent of how usable the new credential is to the people who must adopt it, are shown to generate written and shared workarounds rather than remove the need for them [1], workarounds a compliance-budget account frames as the predictable output of employees weighing a policy's cost against its perceived benefit on each occasion it applies [2], and which an organisation's own security team is shown to see

only a fraction of, however carefully it monitors compliance [3].

Almost none of this literature has followed the credential outside an employer's compliance apparatus. A shared keypad code at a rented share house's front door is rotated on a schedule set by a letting agent or a housing cooperative's own bylaws, with no security team, audit, or line manager positioned to notice when the new code turns up somewhere the tenancy agreement never authorised. The intervention such settings do have available — a reminder, delivered more or less insistently until the point where it is presumed to have worked — borrows its logic from workplace security-awareness training and from nudge interventions shown to raise completion of a specific, dated action elsewhere [4], without the settings' informal, low-oversight, high-turnover character having been tested against it.

This paper reports a fourteen-week field deployment of a lightweight code-rotation reminder tool across a cluster of share houses, compared against each house's existing paper-notice rotation process. Its contributions:

1. We report, to our knowledge, the first field test of a rotation-reminder intervention against a shared household access code specifically, rather than a workplace credential.
2. We find no detectable difference between the reminder tool and the standard paper notice in how quickly a rotated code turns up as informal written notation near the door.
3. We show, via a notification-frequency ablation, that this null result is not an artefact of an under-powered nudge: doubling the reminder's frequency changes the outcome by a matter of hours.

Related work

Workplace security research frames a written or shared workaround not as carelessness but as the visible trace of a policy misfit: an employee's decision to write down or share a credential is shown to follow from how well the credential's rotation and use fit the task it gates, rather than from how clearly the policy was communicated [1], and organisations are shown to see only a fraction of the resulting "shadow" practice, since employees construct it precisely to stay unobtrusive while still getting the job done [3] — a pattern later work argues an organisation should treat as diagnostic signal rather than something to stamp out [5]. A user's decision to route around a piece of security advice is framed, on the same account, as a rational response to a poor cost-benefit trade rather than a lapse [6], a framing later revisited to ask what reference class such a decision should properly be measured against [7].

Reminders and nudges carry their own limits. A warning repeated in the same visual form is shown, via fMRI, to lose the brain's attention within a handful of exposures, an effect a warning that visibly changes each time is shown partly to resist [8], and repeated alerting more broadly is documented, across a wide review of security-operations settings, as degrading rather than sustaining a recipient's responsiveness the longer it continues [9]. An authentication step added for security is shown, in inter-

views with online banking customers, to generate its own friction and workaround talk largely independent of whether it improves security [10], and a behaviour repeated in a stable context is shown to automatise only gradually, along a curve that plateaus at a median of 66 days across a range of everyday behaviours [11] — well beyond the fortnight this paper's own reminder schedule allows. Set against that horizon, a prompt to commit to a specific date and time for a one-off action is nonetheless shown to lift completion measurably above an unprompted control in a workplace field trial [4] — a result this paper's own reminder tool borrows the logic of without importing its one-off framing.

The largest available field measurement of everyday password behaviour, gathered from half a million users over three months, finds reuse and low effective strength the norm rather than the exception [12], a baseline against which any single intervention aimed at one credential's handling has to be read. Sharing a household credential is documented elsewhere as a considered social practice bound up in trust and convenience rather than a lapse in judgement [13], and residents of shared, non-family households are shown to negotiate device and credential access through informal, ad hoc arrangements distinct from the single-owner model most consumer security guidance assumes [14]. Prior work at this institution similarly found a share house's written incident register changing what a recurring argument cited without changing how often the argument itself began [15], and a shared-house parcel locker's own status badge lagging the system's true state by a margin that widens exactly when demand on the shared resource is highest [16].

Method

Recruitment and rotation

Forty-two share houses drawn from a residential letting cooperative's participating listings, each with a shared electronic keypad governing the front door and no other access-control system, were enrolled for a single fourteen-week term. Houses averaged 4.3 residents (SD 1.1) and rotated their keypad code on the cooperative's existing quarterly schedule, which fell two or three times within the study window depending

on a house’s enrolment date (121 rotation events in total, median three per house; Table 1).

Houses were randomly assigned to the reminder-tool arm (n = 21) or to continue the cooperative’s standard paper-notice process (n = 21, control). Within the reminder-tool arm, houses were further split into a daily-notification sub-arm (n = 11) and a weekly-notification sub-arm (n = 10) for the ablation reported below. Because a share house’s residents do not encounter its entry area independently of one another, the house rather than the individual resident is treated as the unit of assignment throughout, consistent with survey evidence that household members’ own contact patterns are highly structured rather than uniformly random [17].

The reminder tool

The tool sends a push notification and a text message to every resident of an enrolled house at T-3 days, T-1 day, and on the day of a scheduled rotation, then continues at the sub-arm’s assigned cadence (daily or weekly) for the following two weeks, prompting residents to check the new code from the app rather than write it down. A resident could dismiss a notification from the lock screen without opening the app; dismissal is not distinguished from reading below, only receipt.

Outcome measure

The outcome of interest is the time from a rotation event to the first legible informal notation of the new code appearing anywhere in the entry area — a sticky note, a whiteboard scrawl, a taped index card — rather than staying in a resident’s memory or the tool’s own interface. A research assistant, blind to arm assignment, photographed each house’s entry area weekly during a scheduled recycling-bin collection visit already built into the cooperative’s routine, and two independent coders scored each photograph for legible notation of the current code (Cohen’s κ = 0.91). Time to first notation for rotation event i is the smallest audited day t at which notation was present:

$$T_i = \min\{t : \text{notation present at house visit day } t\}$$

Self-reported recall

Residents in the reminder-tool arm completed a two-item survey at the end of term recording whether they recalled receiving at least one notification and, if so, how many they estimated seeing.

Results

Arm	Houses	Rotation events
Reminder tool — daily	11	32
Reminder tool — weekly	10	29
Standard notice	21	60

Table 1: Houses and audited rotation events by study arm.

Across the 121 audited rotation events, median time to first informal notation was 2.6 days in the reminder-tool arm (IQR 1.4-4.1) against 2.9 days under the standard paper notice (IQR 1.6-4.4), a difference that did not reach the study’s pre-registered $p < .05$ threshold (Mann-Whitney U , $p = .61$; Figure 1). The two distributions overlap almost completely across their full range, with the reminder tool’s fastest and slowest observed events both inside the standard notice’s own interquartile range.

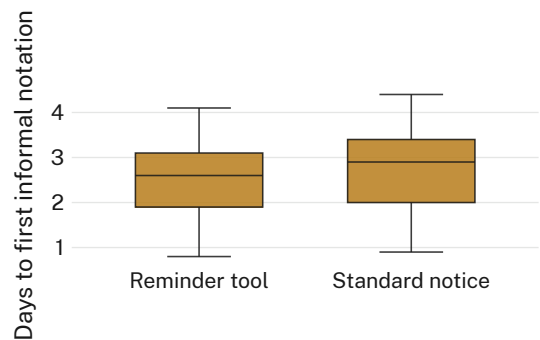


Figure 1: Days from a rotation event to the first legible informal notation of the new code, by study arm. The two distributions overlap almost completely.

Of the 68 informal notations recorded, 61% were a paper sticky note near the keypad, 24% a whiteboard scrawl elsewhere in the entry area,

and 15% a taped index card; the arm a house was assigned to did not predict which of the three a resident reached for ($\chi^2(2, N = 68) = 1.14, p = .57$).

The notification-frequency ablation, splitting the reminder-tool arm into its daily and weekly sub-arms, changes almost nothing (Figure 2): median time to first notation was 2.5 days under daily notifications against 2.7 days under weekly notifications overall ($p = .74$), and the gap stays within two to three hours across every stage of term we can distinguish, from the first five weeks (2.4 versus 2.6 days) through the last four (2.5 versus 2.7 days).

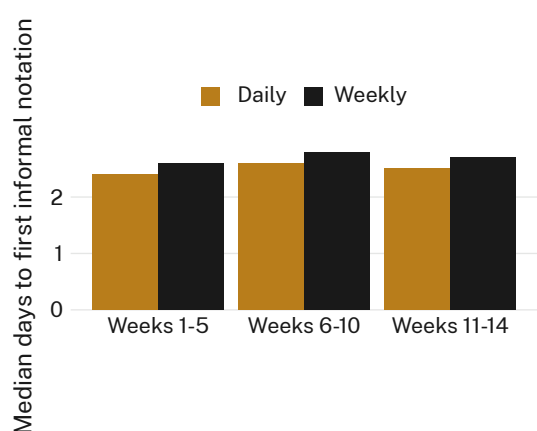


Figure 2: Median days to first informal notation by notification frequency, across three term-stage bins within the reminder-tool arm. Doubling notification frequency changes almost nothing, at any stage of term.

Self-reported notification recall was high: 88% of reminder-tool-arm residents recalled receiving at least one notification, and a median respondent estimated seeing four. Recall was not correlated with how quickly their house's code appeared as informal notation ($r = 0.04$, n.s.) — a resident who reported ignoring every notification was, on this evidence, no more likely to reach for a sticky note than one who reported reading every one.

Discussion

Two independent checks agree that this reminder tool changed almost nothing about how a rotated share-house door code reaches paper. The headline comparison finds no detectable difference between a resident nudged by push notification and text message and a resident relying on the cooperative's existing paper notice, and the frequency ablation rules out an under-

powered nudge as the explanation: residents notified daily wrote the code down on essentially the same timeline as residents notified weekly. High self-reported recall, uncorrelated with the outcome the tool was meant to move, suggests the notifications were seen and set aside rather than unseen and missed.

We read this as evidence that a code's informal legibility is set by a share house's own coordination needs — a departing housemate, a visiting friend, a delivery driver who needs the gate code read out over an intercom — rather than by how often, or how insistently, residents are individually told to remember it themselves [2], [3]. Seeding the same information through whichever housemate a household would itself nominate as the one who tells everyone things is shown, in other field settings, to diffuse further than the identical information pushed to each person individually [18], a mechanism this deployment's house-level notifications did not attempt to use and cannot rule out as the households' own, faster channel. A workplace's compliance apparatus can, at least in principle, monitor and sanction the written workaround it produces; a share house has no such apparatus, and this tool's own notifications joined, rather than replaced, the household's existing informal channel.

Limitations

This deployment covers one letting cooperative's participating houses across a single term; a longer window, or a cooperative with stricter code-complexity requirements, might surface a reminder effect this study's audit was not positioned to detect. The weekly audit cadence caps the outcome's resolution at roughly plus or minus three days, which, if anything, works against finding a real difference between arms rather than manufacturing a spurious one, so the null result reported here is if anything conservative. Coders were blind to arm assignment but not to the study's purpose, a limitation shared with almost every audit of a household's own informal behaviour that does not install a camera nobody consented to.

Conclusion

A lightweight reminder tool, pushing notifications to every resident of a share house at increasing frequency around a scheduled door-

code rotation, did not measurably change how quickly the new code appeared as a sticky note, a whiteboard scrawl, or a taped index card near the keypad, and doubling the reminder's frequency moved the outcome by hours rather than days. Self-reported recall of the notifications was high and uncorrelated with the behaviour they were meant to prevent. Where a workplace's compliance apparatus can monitor and sanction the written workaround a poorly-fitted policy produces, a share house has none, and this deployment's own notifications appear to have joined its informal channel rather than displaced it. Future work might test whether a reminder aimed at the household's coordination need, rather than at the individual resident, fares any better.

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